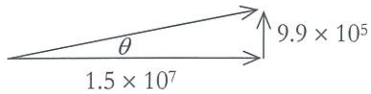
21. (e) Use vectors

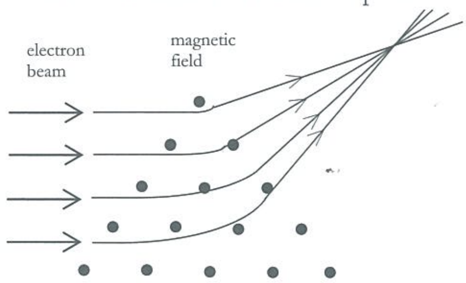


$$\tan\theta = \frac{9.9\times10^{5}}{1.5\times10^{7}}$$

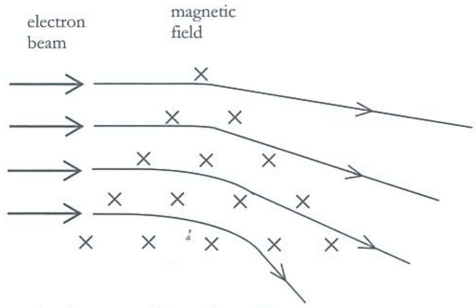
$$\therefore \theta = 3.78^{\circ}$$

(f) Electrons would describe a circular path and hence the direction of the force acting on them is continually changing.

22. (a) upwards
(b) the bottom part
(c) the beam will tend to focus to a point.



(d)



The beam will tend to diverge.

23. (a) upwards
(b) for no nett force
$qvB = mg$

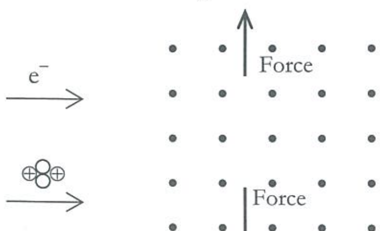
$$\therefore v = \frac{mg}{qB}$$

$$= \frac{(9.11\times10^{-31})(9.80)}{(1.6\times10^{-19})(2.38\times10^{-5})}$$

$$= 2.34\times10^{-6}\,\text{ms}^{-1}$$

(c) Since most electrons are moving with far greater speeds than 2.34×10^{-6} the effect of the magnetic field would be the greatest.

24. (a)



(b) Since $F = q\,v\,B$, the force on the alpha particle (× 2) will be twice as great.

24. (c) $a = \dfrac{F}{m}$ Since mass of the electron is far less than that of the alpha particle, it will have the greatest acceleration.

25. (a) $$r = \frac{m\,v}{Bq} = \frac{(1.67\times10^{-27})(8.0\times10^{4})}{(1.6\times10^{-19})(4.5\times10^{-2})}$$

$$= 1.85\times10^{-2}\,\text{m}$$

(b) $r \,\alpha\, \dfrac{m}{q}$

$$\therefore r_{\alpha} = 1.85\times10^{-2}\times\frac{4}{2}$$

$$= 3.70\times10^{-2}\,\text{m}$$

(c) $\dfrac{r_{\alpha}}{r_{p}} = 2$

26. Since $r = \dfrac{m\,v}{Bq}$ $\therefore m = \dfrac{rBq}{v}$

$$\therefore m = \frac{(9.40\times10^{-2})(5.5\times10^{-2})(2\times1.6\times10^{-19})}{1.71\times10^{4}}$$

$$= 9.67\times10^{-26}\,\text{kg}$$

TRIAL TEST 1 – SOUND WAVES

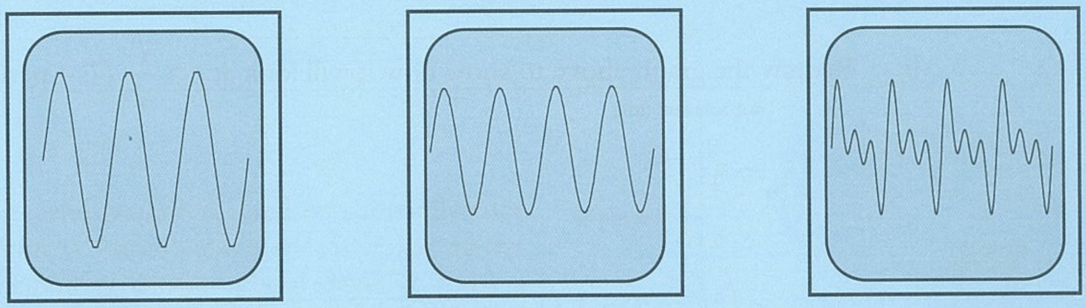
- Time allowed: 60 minutes
- Total marks: 100%

Section A - 30% (30 marks)
Section B - 50% (50 marks)
Section C - 20% (20 marks)

- Physical data: Use (1) velocity of sound in air $= 3.40 \times 10^2$ ms^{-1}
  velocity of sound in seawater $= 1.53 \times 10^3$ ms^{-1}
  (2) formulae and constants, as may be necessary, from pages 2-6 at the front of the text book.

Section A

1. Three different sounds *A*, *B* and *C* were played into a microphone connected to a cathode ray oscilloscope (CRO). Their individual traces are shown below. Assume similar settings of the CRO for each sound.



SOUND *A* SOUND *B* SOUND *C*

Describe the difference/s that a listener would notice between

(a) Sounds *A* and *B* ______________________________

____

______________________________ (1)

(b) Sounds *B* and *C* ______________________________

____

______________________________ (1)

2. Under similar conditions sound waves of all frequencies travel at the same speed. From your knowledge of the nature of sound explain clearly why this statement must be correct.

____

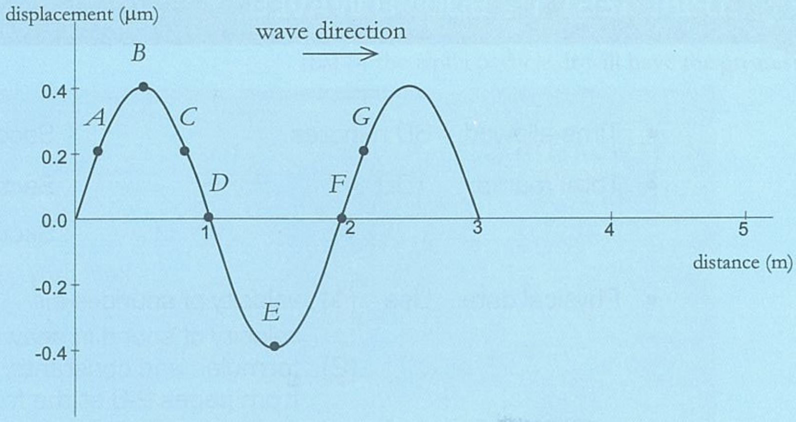
____

____

____

______________________________ (2)

3. The graph below shows the displacement of air molecules from their mean position at an instant in time, $t = 0$.



(a) From the graph determine the following:

(i) amplitude ____________________

(ii) wavelength ____________________

(iii) two points in phase ____________________

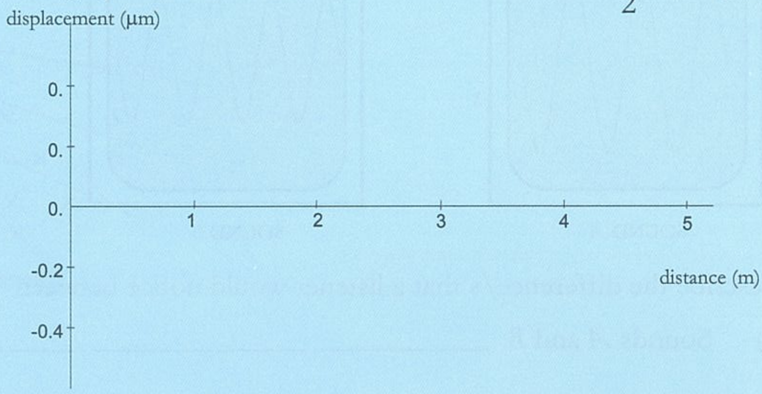
(iv) two stationary points ____________________

(v) a point moving away from its mean position ____________________

(vi) two points moving with greatest speed ____________________

(3)

(b) Redraw the graph above to show how it will look at $t = \frac{T}{2}$ (T = period).



(3)

4. Echo sounding devices are often used to determine the depth of oceans and rivers. These devices emit short pulses of ultrasound (about 50 kHz) towards the ocean floor and then detect the reflected sound.

(a) Explain how it is possible to measure ocean depth in this way.

____

____

(2)

(b) Why are such high frequency sounds used for this purpose?

____

____

(1)

(c) If the delay time between sounding and receiving a signal is 110 ms what is the depth of the water? (You may need to refer to physical data tables.)

____

____

____

(2)

5. A loud sound travelling in cool air, encounters a region of warm air. Indicate briefly how each of the following may be affected as the sound passes through to the warm air and beyond.

(a) speed ______________________ (d) wavelength ______________________

(b) direction ______________________ (e) intensity ______________________

(c) frequency ______________________

(5)

6. Frank and Robbie are carrying out sound experiments in the laboratory and each sets up a sound generator to an amplifier and loudspeaker. They each set their sound generators to 1000 Hz but find that when the speakers are sounded together the sound heard continuously flutters in intensity.

(a) What is the likely cause of this effect? Explain clearly.

____

____

(2)

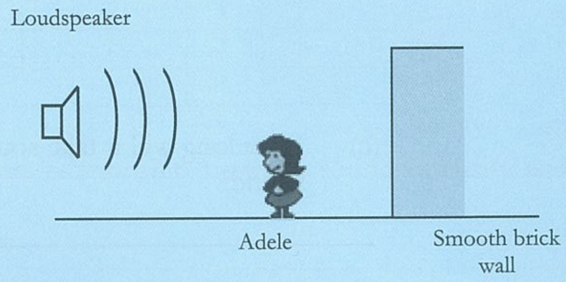
(b) What can the two experimenters do to eliminate this effect?

____

____

(2)

7. A loudspeaker located a short distance from a smooth brick wall is sounding the beginning of the school day. It emits a loud single frequency sound of 850 Hz. Adele notices that as she walks between the speaker and the wall, she comes across 'dead spots' where there is very little sound.



(a) Explain clearly what causes this phenomenon.

____

____

____

____

(3)

(b) What name is given to these 'dead spots'?

____

(1)

(c) Determine the likely distance between consecutive 'dead spots'

____

____

____

(2)

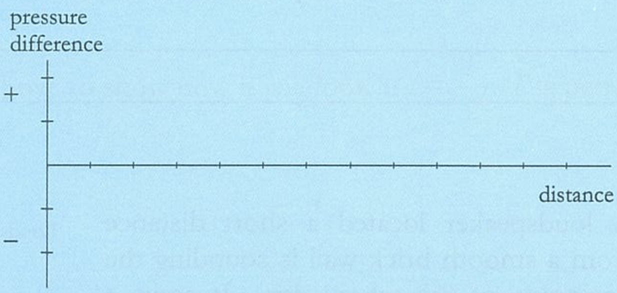
Section B

8. A loudspeaker emits a sound of 1.00 kHz towards a microphone as shown. The microphone is 2.0 m from the speaker.

(a) Determine the wavelength of the sound being emitted.

(2)

(b) Complete the graph below to show how air pressure varies with distance from the microphone 4.0 ms after the commencement of the sound. Assume that initially a compression wave leaves the cone. Pressure levels are arbitrary. Select appropriate distance units.

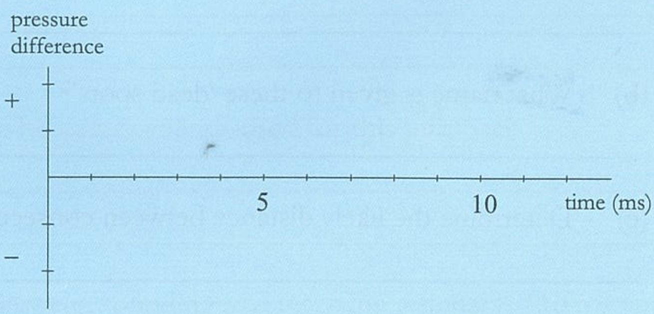


(6)

(c) How long will it take sound to reach the microphone after the commencement of the sound?

(3)

(d) Sketch a pressure difference versus time graph as would be recorded at the microphone for the first 0.010 seconds of sound emitted by the speaker.



(6)

9. Paula and Melissa carried out an experiment using a sound level meter to take readings of the sound from a CD player. Their results for different distances are shown below.

Distance from CD player (m)	2	4	6	8	10	12
Sound level (dB)	90	84	80.5	78	76	72
Sound Intensity (Wm^{-2})						

(a) Complete the table by calculating the sound intensities at the different distances. (6)

(b) By what factor did the intensity of the sound change between:

(i) 2 m and 4 m

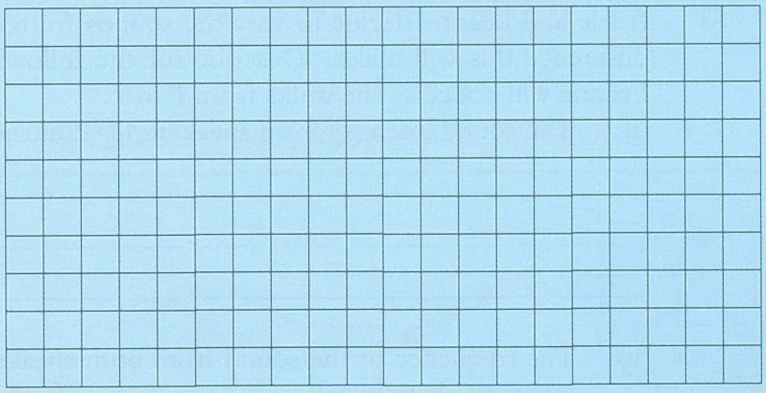
____

(ii) 4 m and 8 m

____

(2)

(c) Use the data to construct a graph of sound intensity versus distance from the source.



(6)

(d) Using the graph (or data) determine the relationship between sound intensity and distance of a sound from the source.

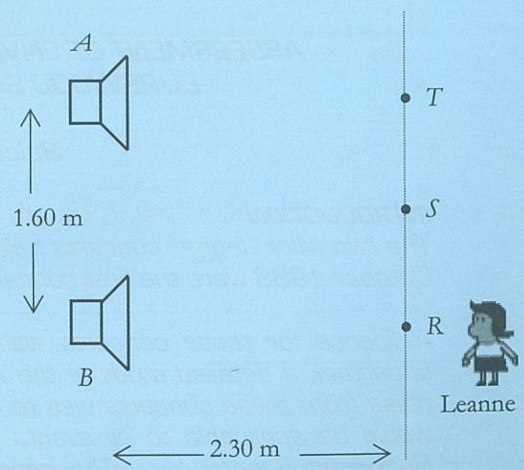
____

____

____

(3)

10. Leanne and Adele connect two similar speakers (A and B) to a sound frequency generator so that each speaker will emit sounds that are in phase and of equal frequency and intensity. They investigate the loudness of the sound produced at points along a line parallel to the speakers. They use a frequency of 685 Hz. Dimensions of the layout of the experiment are shown on the diagram. Leanne walks along the line AC and notices that maximum sound intensities occur at R and T (directly in front of the speakers) and at S (a point equidistant from each speaker). Quiet spots are noticed in between.



10. (a) What is the cause of this effect? Explain clearly.

(3)

(b) From the diagram determine the difference in the distance between AT and BT.

(4)

(c) Use your result in (b) to determine the velocity of sound during this experiment.

(3)

(d) Adele and Leanne decide to vary the sounds from each of the speakers to see what difference this will make. Describe for the following situations the difference that Leanne will notice as she walks from P to T.

(i) The sound intensity from speaker A is doubled while B remains the same.

(3)

(ii) The frequency of the sound from both speakers is doubled (1370 Hz) and the intensities kept equal.

(3)

Section C

Read the following article, then answer question 11.

ASSESSMENT OF ENVIRONMENTAL NOISE - MICK JAGGER CONCERTS BURSWOOD SUPERDOME, 10 AND 11 OCTOBER 1988

Selected extracts from EPA report.

INTRODUCTION

The two Mick Jagger concerts held at the Burswood Superdome on Monday 10 and Tuesday 11 October 1988 were the first concerts by a major rock music artist to be held at this venue.

A licence for these concerts was issued by the Environmental Protection Authority who also undertook a detailed study of the noise emissions generated. One of the reasons for assessing noise from these concerts was to establish an effective noise emission level control strategy that would be applicable to all events having a potential for high noise emission levels held at the Burswood Superdome. The second major reason for the assessment was to establish the environmental impact of noise from these two concerts and associated activities on residents living in the vicinity of the Superdome.

To provide data for this assessment, noise levels inside the Superdome during the event were recorded and the noise environments at selected locations in the residential areas around the venue were measured and subjectively assessed. Measurements in the residential areas were carried out over period of thirteen days or more to allow comparison of the noise situation on 10 and 11 October, when the two concerts were held, with other days when there was not a major event at the Superdome.

NOISE ENVIRONMENT IN NEARBY AREAS

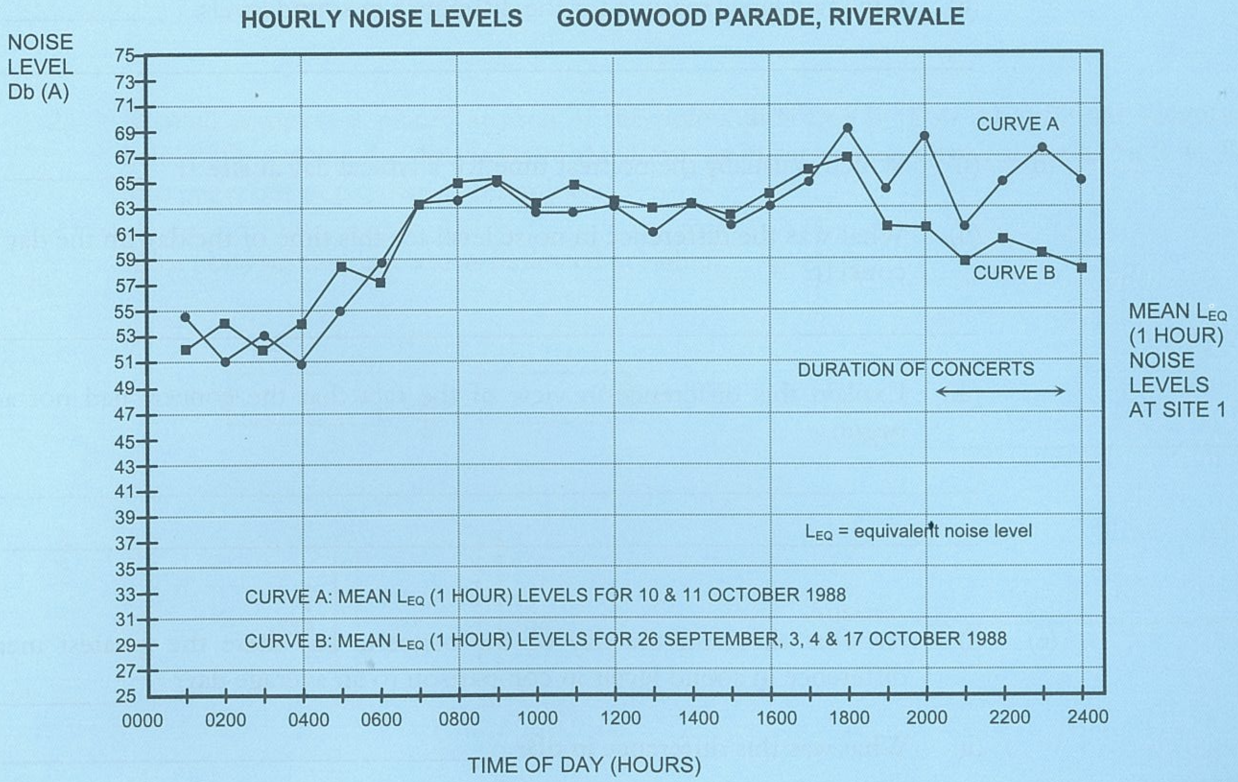
Noise measurements were made over periods of thirteen (13) to twenty-eight (28) days which included 10 and 11 October, at six residential sites in the general vicinity of the Superdome. The closest site to the Superdome was at Goodwood Parade Rivervale, adjacent to Rivervale Railway Station. The graphed results refer to this site.

For each of the six sites, noise levels in the absence of a function at the Superdome were established by selecting four typical days and calculating the average hourly noise levels. Weather conditions, the availability of data and evidence of other activities affecting noise levels were all considered when selecting the particular days.

The graph shows

a) the average equivalent steady noise level (L_{EQ}) for each hour of the two concert days;

b) the average equivalent steady noise level (L_{EQ}) for each hour of those days selected as the control bases.



11. (a) What were the two major reasons for the EPA undertaking a study of the noise levels from the Mick Jagger concerts?

____

____

____

____

(4)

(b) Measurements of noise levels at the six chosen residential sites surrounding the Superdome were taken over a period of several days. Explain:

(i) why these readings were necessary; ______________________

____

(ii) why readings were taken over several days; ______________________

____

(iii) how the readings were averaged. ______________________

____

(3)

Refer to the graph of hourly noise levels at Goodwood Parade, Rivervale (site 1) for the following.

(c) (i) Estimate the average noise level at site 1 for the hours:

- 12:00 AM to 4:00 AM ______________________
- 8:00 AM to 12:00 PM ______________________

(ii) Give two likely reasons for the difference in sound levels. ______________

____

(4)

(d) (i) Which is usually the noisiest time for a typical day at site 1? ______________

(ii) What was the difference in noise level for this time of the day on the day of the concert?

____

(iii) Explain this difference in view of the fact that the concert had not actually begun.

____

____

(3)

(e) (i) At what time, on the day of the concert, was there the greatest measured difference in sound levels in comparison to an average day? ______________

(ii) What was this difference in dB? ______________________

(iii) Use your answer in (b) to calculate the factor by which the actual intensities (energies) of the sounds on the two days differ.

____

____

____

(6)

TOTAL 100 MARKS

TRIAL TEST 2 – ELECTRIC POWER

- Time allowed: 60 minutes
- Total marks: 100%

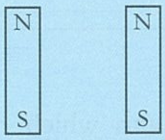
Section A - 30% (30 marks)
Section B - 50% (50 marks)
Section C - 20% (20 marks)

- Physical data and formulae: Use formulae and constants listed on pages 2-6 of this textbook.

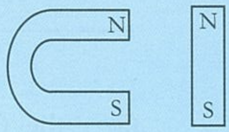
Section A

1. Sketch the magnetic fields surrounding the magnets in the situations shown below. Assume all magnets are of equal strength.

(a)



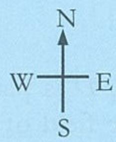
(b)



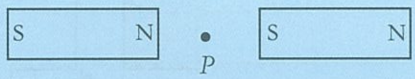
(4)

2. A small compass is placed at each of the points indicated by *P* in the different situations below. Determine the directions of the compass in each case. Give answer as N, S, E, W, into page, out of page or directionless.

Directions on the page are as indicated at right.

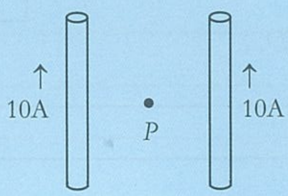


(a)



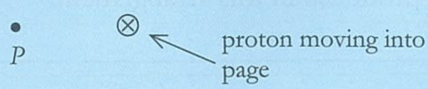
Answer ______________________

(b)



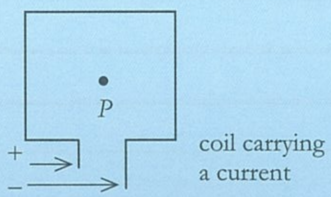
Answer ______________________

(c)



Answer ______________________

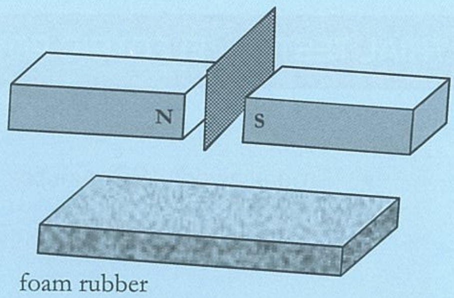
(d)



Answer ______________________

(4)

3. A flat sheet of copper metal is allowed to fall freely between and beyond two strong bar magnets. It does not touch the magnets and is allowed to fall onto soft foam rubber.



(a) After repeating this procedure several times, it is noticed that the copper metal becomes warm? Explain the cause of this heat.

____

____

(2)

(b) How would the amount of heat produced in the metal sheet be affected if

(i) the magnets are moved slightly closer together (assume all other conditions remain the same)? Why?

____

____

(2)

(ii) the flat sheet was made of aluminium which has a higher resistivity than copper? Why?

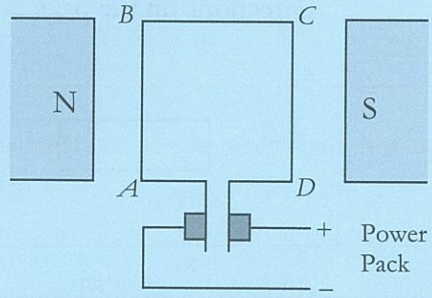
____

____

(2)

4. A coil A, B, C, D is placed as shown in a strong magnetic field of 0.55 T. A current of 3.60 A passes through the coil from the power pack.

Assume $AB = CD = 50$ cm,
$BC = AD = 40$ cm.



(a) Determine the magnitude and direction of the force exerted on CD.

____

____

____

(2)

(b) Calculate the total torque produced in this arrangement.

____

____

(2)

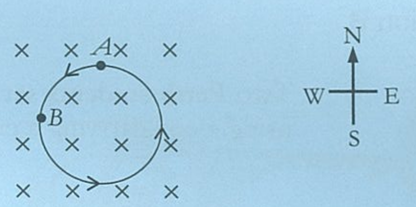
(c) State two means by which this torque could be increased.

____

____

(2)

5. The diagram at right shows a charged particle moving in a circular motion in a magnetic field. The field is into the page.



(a) Is the particle likely to be a proton or an electron?

______________________________ (1)

(b) What is the direction of the force on the particle at points *A* and *B*? (N, S, W, or W)

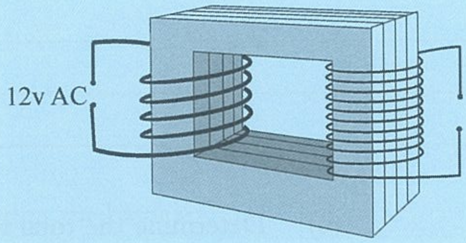
(i) At *A* ______________ (ii) At *B* ______________ (2)

(c) How would the path change if the particle had double the charge (assume all other conditions unchanged). Why?

____

______________________________ (2)

6. A typical transformer is shown at right. It has about twice as many turns of wire in the secondary coil than it has in the primary coil.



(a) Will the induced voltage be greater or smaller than the applied voltage? Explain.

____

____

______________________________ (2)

(b) Explain why the secondary coil is made of thinner wire.

____

____

______________________________ (1)

(c) Why is the core of the transformer laminated?

____

______________________________ (1)

(d) Will the transformer work equally well with DC current? Why/Why not?

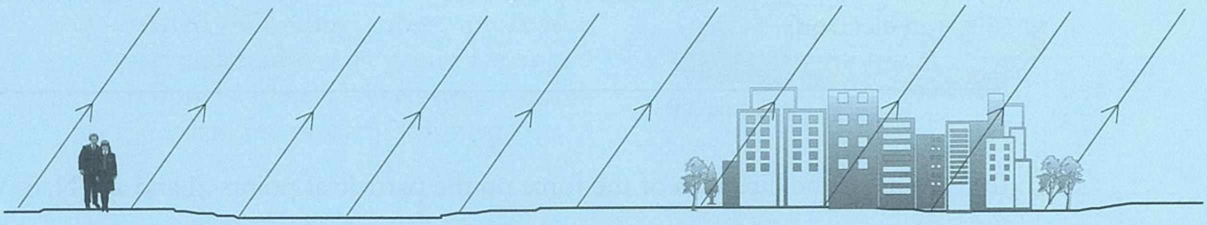
____

____

______________________________ (1)

Section B

7. Two Perth students set out to investigate the possibility of creating a simple electric motor using a coil carrying a current in the earth's magnetic field.



Assume the earth's magnetic field where the students are is 5.90×10^{-5} T at 66° to the horizontal. The students construct a single copper coil 50 cm by 50 cm and arrange it so that it is freely pivoted.

(a) How should the students arrange their coil to achieve maximum torque? Include a clearly labelled sketch showing orientation of coil, current flow, forces produced and any other relevant information.

(6)

(b) Determine the total torque on the coil that could be achieved in this manner with a current of 5.0 A.

(4)

(c) Discuss the likely outcome or results of the students experiment. Give reasons.

(2)

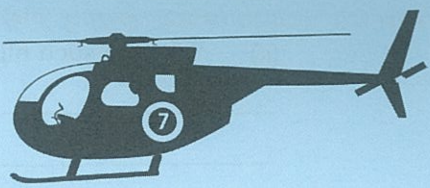
(d) How could the boys improve the torque produced by their simple motor?

(3)

(e) Could this experiment give better results at some other point on earth? Explain.

(2)

8. A Channel 7 helicopter with blades that span 10.7 m across is operating in an area where the earth's magnetic field is 5.85×10^{-5} T at an angle of 66° to the horizontal. Prior to take off the rotor blades are moving at 375 rpm in a clockwise direction when viewed from the ground.
It is found that the motion of the blades causes a voltage difference within them.



(a) Which part of the rotor blades

(i) would be positive? ______________________

(ii) would be negative? ______________________

(2)

(b) Calculate the vertical component of the earth's magnetic field at this point.

(3)

(c) Calculate the total flux enclosed in the area swept out by the blades.

(3)

(d) What would be the maximum EMF induced in the blades?

(5)

(e) Between which points in the blades would this maximum EMF exist? Why?

(2)

(f) The helicopter, which is facing south, tends to lean forward as it takes off, that is its tail lifts. How will this affect the induced EMF? Why? (Assume blades still moving at 375 rpm).

(2)

9. The Muja power station in the South West of W.A. provides electricity to Kalgoorlie by way of a 220 kV transmission line 650 km in length.

(a) Why are such high voltages used to transmit electricity over this distance?

(2)

(b) What are some of the difficulties and possible dangers of transmitting power at such high voltages?

(2)

(c) If this line is transmitting 20 MW of power, determine:

(i) current on the line;

(ii) power loss due to resistive heating if the total resistance of the line is 500Ω.

(8)

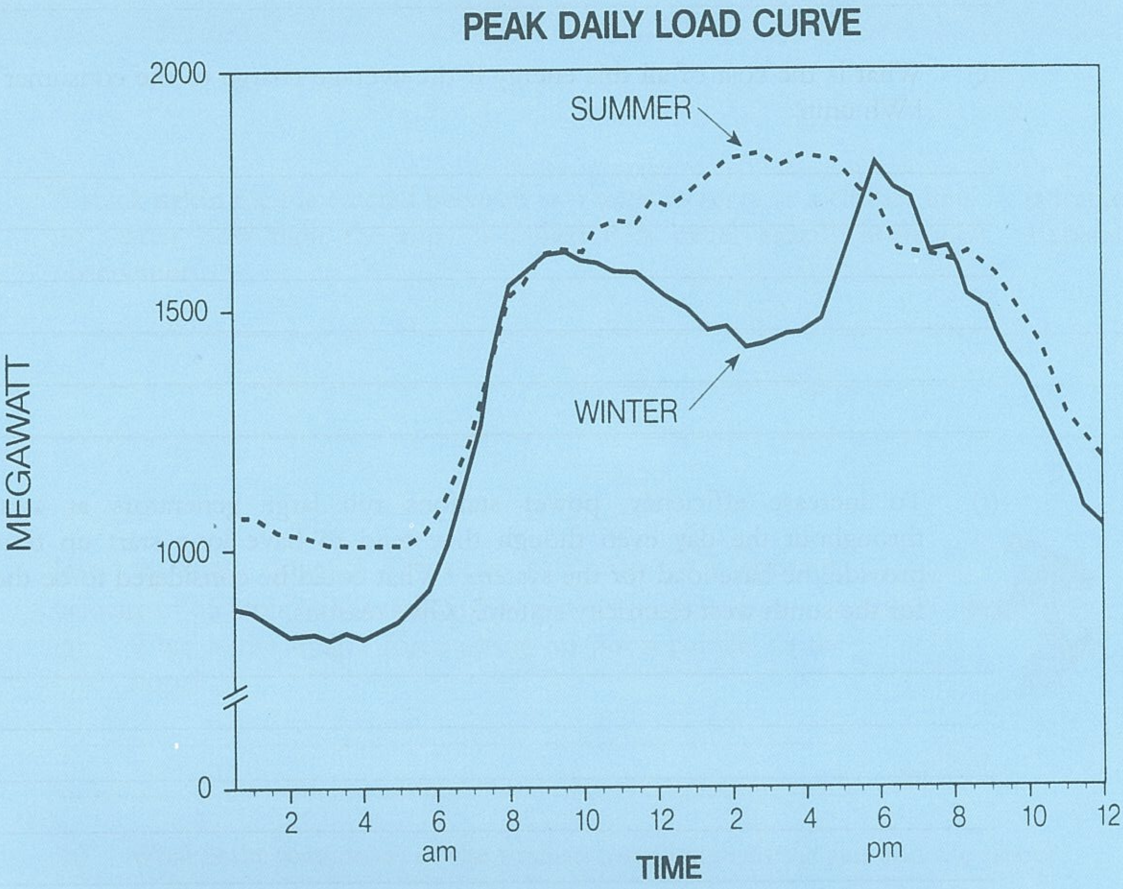
(d) Determine the power loss if the power had been transmitted from Muja to Kalgoorlie with a 66 kV line. Comment on your result.

(6)

Section C

10. The graph below shows the seasonal variation in daily power consumption for a local electricity system.

Use the graph to answer the following questions.



(a) What is the highest rate of power consumption during a summer day?

____ (2)

(b) At what time of the day is the greatest amount of electricity being used:

(i) in summer? ________________ (ii) in winter? ________________

(2)

(c) Give likely reasons as to why your two answers to (b) are different.

____

____

____

____ (4)

(d) Estimate the amount of electrical energy consumed on a summer day between 8.00 am and 8.00 pm. Give your answer in kWh.

(5)

(e) What is the cost of all this energy if the average charge to the consumer is 12.7 ¢ a kWh unit?

(5)

(f) To increase efficiency, power stations run large generators at a steady rate throughout the day even though they tend to have long start up times. These provide the base load for the system. What could be considered to be the base load for the south west electricity system? Give reasons.

(2)

TOTAL 100 MARKS